Section 2

Group: Sebastián Encalada, Juan Diego Chicaiza, María Emilia Granda.

① Modular Arithmetic

1.1 $x \equiv 457 \pmod{13}$

$$\begin{array}{r|l} 457 & 13 \\ \hline 67 & 35 \\ \boxed{2} & \end{array}$$

$x = 2 \pmod{13}$

1.2 $x = -25 \pmod 7$

$$\begin{array}{r|l} 25 & 7 \\ \hline 4 & 3 \end{array}$$

$-25 \equiv -4 \pmod 7$

$-4+7 = 3$

$x = 3 \pmod 7$

② Euclidean Algorithm

2.1 $\gcd(1160, 480) = 40$

$1160 = 480(2) + 200$

$480 = 200(2) + 80$

$200 = 80(2) + 40$

$80 = 40(2) + 0$

2.2 $1160x + 480y = \gcd(1160, 480)$ $\quad x = 5,\ y = -12$

$1160(5) + 480(-12) = 40$

$40 = 200 - 80(2)$

$80 = 480 - 200(2)$

$40 = 200 - 2(480 - 200(2))$

$40 = 200 - 2(480) + 4(200)$

$40 = 5(200) - 2(480)$

$200 = 1160 - 480(2)$

$40 = 5(1160 - 2(480)) - 2(480)$

$40 = 5(1160) - 10(480) - 2(480)$

$40 = 5(1160) - 12(480)$

③ Euler's Totient Function

3.1 $\phi(21)$

$21 = 7 \quad 3$

$\phi(n) = n\left(1 - \frac{1}{P_1}\right)\left(1 - \frac{1}{P_2}\right)$

$\phi(21) = 21\left(1 - \frac{1}{7}\right)\left(1 - \frac{1}{3}\right)$

$\phi(21) = 21\left(\frac{6}{7}\right)\left(\frac{2}{3}\right)$

$\phi(21) = 12$

3.2 $\phi(17) = 17 - 1$

$\phi(17) = 16$

3.3 $\phi(360)$

$360 = 36(10)$

$= (2^2)(3^2)(2)(5)$

$= (2^3)(3^2)(5)$

$\phi(360) = 360\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)$

$\phi(360) = 360\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{4}{5}\right)$

$\phi(360) = 96$

4 Information Theory

$P(A) = \frac{1}{2}$ $\quad P(B) = \frac{1}{4}$ $\quad P(C) = \frac{1}{8}$ $\quad P(D) = \frac{1}{8}$

$$H(X) = \sum_i p_i \log_2(p_i)$$

$$H(X) = -\left(\frac{1}{2}\log_2\left(\frac{1}{2}\right) \; \frac{1}{4}\log_2\left(\frac{1}{4}\right) \; \frac{1}{8}\log_2\left(\frac{1}{8}\right) \; \frac{1}{8}\log_2\left(\frac{1}{8}\right)\right)$$

$$H(X) = -\left(\frac{1}{2}(-1) + \frac{1}{4}(-2) + \frac{1}{8}(-3) + \frac{1}{8}(-3)\right)$$

$$H(X) = -(-1 - 3/4)$$
$$= 7/4$$
$$= 1.75 \text{ bits}$$

5 Bayes' Theorem

$P(A) = 0.001$

$P(F|A) = 0.99$

$P(F|\neg A) = 0.01$

$$P(A|F) = \frac{P(F|A)\,P(A)}{P(F|A)\,P(A) + P(F|\neg A)\,P(\neg A)}$$

$$= \frac{0.99\,(0.001)}{0.99\,(0.001) + (0.01)\,(0.999)}$$

$$= 0.0902 \Rightarrow 9.02\%$$